

Data Management Technologies

Unit 2 - Consolidation Task 3

Part 1

Chapter 4, Questions 4.8 and 4.9.

4.8) Identify the foreign keys in this schema.

Explain how the entity and referential integrity rules apply to these relations.

The foreign keys are as follows for each table:

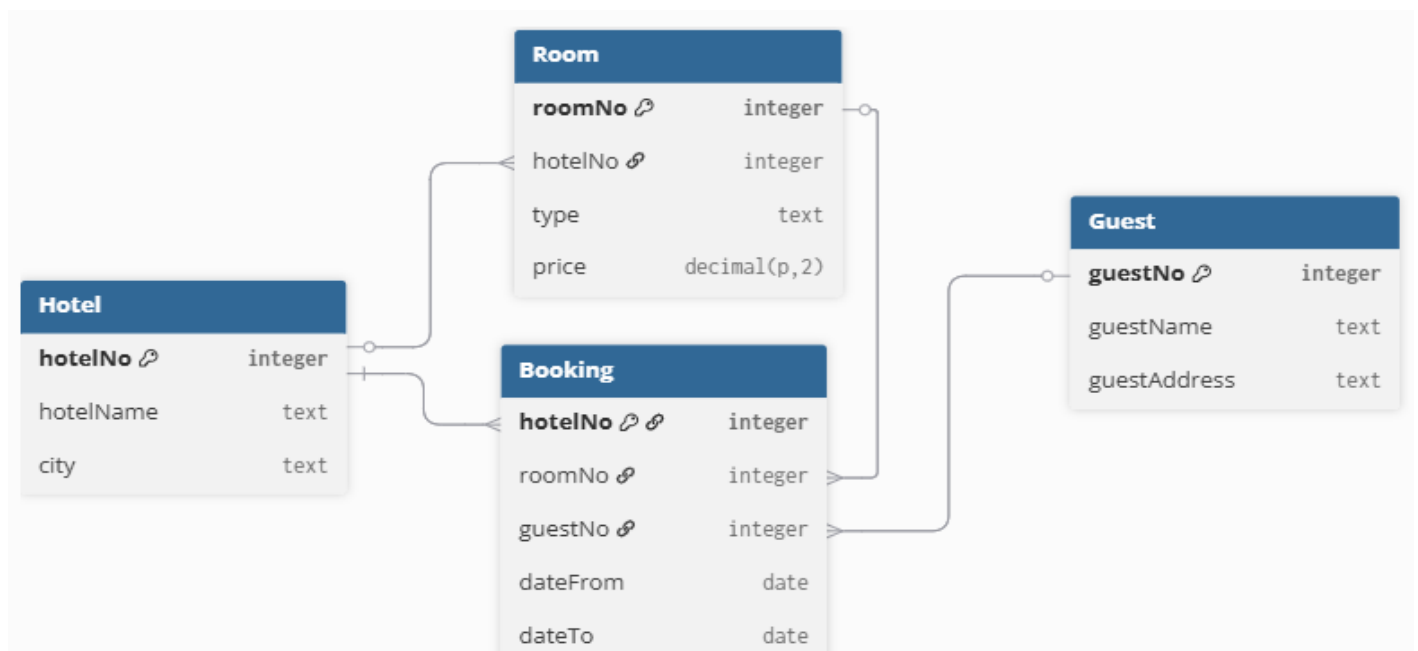
- **Room:** hotelNo
- **Booking:** hotelNo, guestNo, roomNo

Entity integrity rule: A primary key must be unique and not null. In this scenario, this means that roomNo, hotelNo, guestNo and dateFrom must be populated with unique values.

Referential integrity rule: A foreign key must match an existing primary key. The key hotelNo in the Room table must match a record in the hotel table.

The foreign keys in the Booking table must match the respective primary keys in the Hotel, Guest, and Room tables.

4.9) Produce some sample tables for these relations that observe the relational integrity rules. Suggest some general constraints that would be appropriate for this schema.



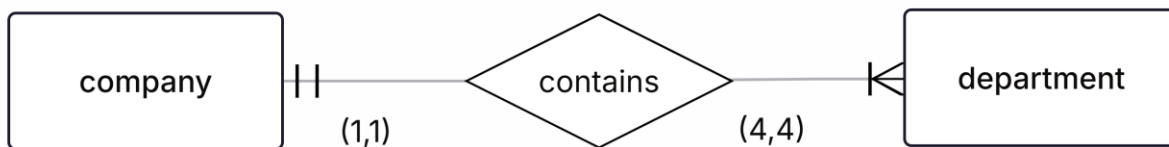
General constraints

- **NOT NULL:** Primary keys (hotelNo, RoomNo, and guestNo) must be populated.
- **UNIQUE:** Primark keys must be unique. The key guestAddress should also be unique.
- **CHECK:** Within Bookings, dateFrom should be in the future. It should be within the next 20 years. Bookings' dateTo should be after the dateFrom.

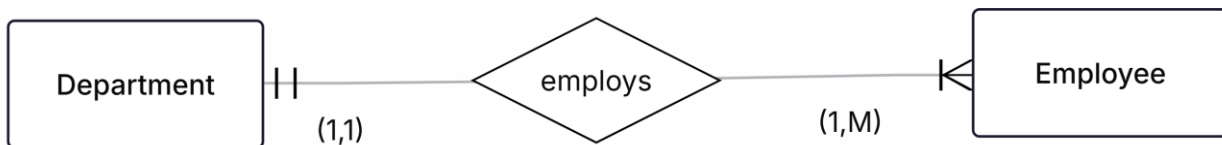
Part 2

Chapter 12, Question 12.10

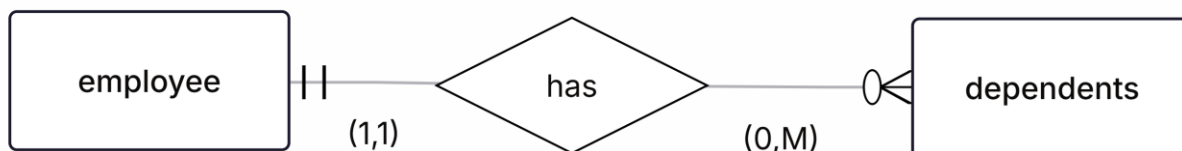
A). Each Company has four departments. Every department is linked to one company.



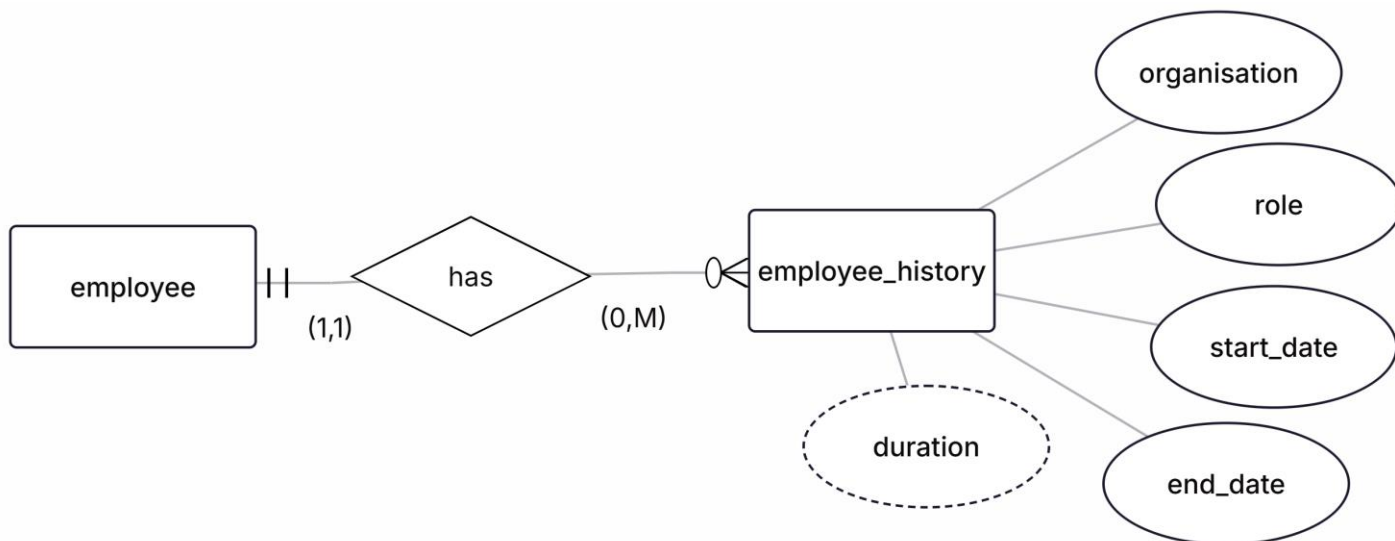
B) Each department has at least one employee. Each employee works for one department.



C) Each employee has none, one or many dependents. Each dependent is linked to an employee.



D) Each employee may or may not have an employment history.



E) Represent all the ER models described in (a), (b), (c), and (d) as a single ER model.

